

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
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Fire and Explosion Risk Assessment Methodology and Assumptions

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Revision History

Rev No.	Date	Description of Revision
R01	08-06-22	Issued for Client's Review
R02		
A01		

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-THS-PRC-001

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEM as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
BLEVE	Boiling Liquid Expanding Vapour Explosion
CAM	Congestion Assessment Method
ER	Emergency Response
ESD	Emergency Shut Down
FERA	Fire and Explosion Risk Assessment
FRED	Fire, Release, Explosion, Dispersion
H&MB	Heat & material Balance
LFL	Lower Flammable Limit
PFP	Passive Fire Protection

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

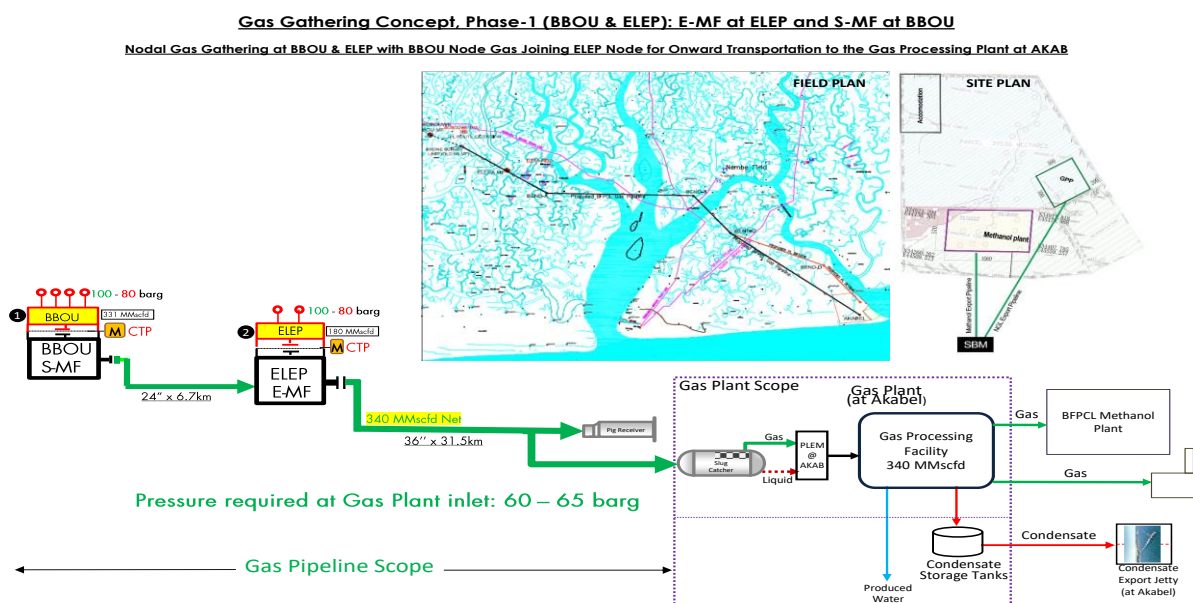


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of the Fire and Explosion Risk Assessment (FERA) is to identify the consequences of ignited and non-ignited hydrocarbon releases. These consequences will then be assessed considering the following:

- Layout & Separation
- Isolation & Blowdown
- Fire & Gas Detection
- Passive Fire Protection
- Active Fire Protection
- Bunding & Drainage

The results of the consequence assessment will be used to assess whether there are any changes that could be made in the above areas to reduce risk, or where there are opportunities to, for example, reduce provision or reduce separation. This document outlines the methodology and assumptions that will be used in conducting the FERA studies

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

Document Number	Document Title
DEP 80.47.10.30 Gen	Assessment of Fire Safety for Onshore Installations
SR13.14005	Shell ALARP Guide

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. METHODOLOGY

3.1. Modelling

First is to carry out Hydrocarbon Release Scenario Event Tree Analysis and estimate the frequencies of the various events. The FRED hazard consequence modelling software will then be used to identify the size of fires (pool & jet) and gas clouds, etc. resulting from hydrocarbon release scenarios events. The Pressurised Release (Known Reservoir Pressure) model will be used. This will give the release rate, fires / thermal radiation distances and dispersion results.

Individual release scenarios will be identified for each isolatable section. These will cover releases from each major equipment item, gas and liquid releases, and releases from a number of different hole sizes.

The following parameters will be used:

Table 1 - Modelling Parameters

Hole Sizes	10mm 25mm 50mm 100mm	
Heat Contours	4.7kW/m ²	Limit for protected personnel to take actions lasting several minutes.
	6.3kW/m ²	Limit for appropriately dressed personnel to take actions lasting up to 1 minute (e.g. escape).
	25kW/m ²	Limit for damage to steel structures
	37.5kW/m ²	Significant damage to steel structures
Gas Dispersion	To LFL	
Wind Conditions	5m/s and 10m/s wind speeds D (neutral) and 2F stability conditions. 0.1m surface roughness	

Releases will be modelled using the initial, pre-ESD, release rate (due to the normal operating pressure).

The pressures used for the initial release rate will be taken from the Heat and Material Balances (H & MB) of the project

The sensitivity of the dispersion results to the wind conditions will be tested. Selected results will be compared with those for 2F (2m/s, very stable) conditions.

3.2. Consequence Assessment

When deciding which strategy to take for a particular fire hazard, the risk associated with the option will be assessed to ensure that asset risk is acceptable and safety risk is ALARP. Where the risk is to the assets, some qualitative cost-benefit analysis will be performed to justify the chosen strategy. The consequence assessment will aim to identify which strategy is the most appropriate and to identify specific prevention or containment measures where appropriate. Control measures will only be recommended where the consequences are likely to be reduced enough to justify the cost of implementing the control.

It will be assumed that ESD and blowdown will function correctly, except in cases where the probability of failure was considered to be particularly high.

The basis of the consequence assessment in each of the key areas is described below.

3.2.1. Prevention Measures

The prevention of leaks and the prevention of ignition will be the primary means of reducing fire hazards. These prevention measures will be applied across the whole project facilities and so will not be discussed for each isolatable section.

Leak sources will be minimised, particularly at the high-pressure gas sections. This will be achieved by the minimisation of valves and flanges, and by robust design and minimisation of small-bore lines and instrument connections.

Flanges are a potential source of leakage, and high pressure, low mol. weight gas has a particularly high leak potential. Flanges present a higher escalation risk, as lines will tend to 'spring' at flanges when exposed to heat.

Procedural controls will also play a key role in preventing releases:

- Permit to Work (JHA)
- Isolation Standards
- Rigging & Lifting Standards

3.2.2. Layout & Separation

Pool Fires

Minimum separation distances between equipment will be based on the extent of the 25kW/m² heat radiation pool fire contour. Various release sizes will be modelled as shown in Table 1 – Modelling parameters of section 3.1 above but only the consequences for 10mm hole sizes will be evaluated as the design case for pool fires. It is considered reasonable to specify drainage and layout such that pools above this size cannot form. The pool centre will be taken as the edge of the module.

Plant exposed to 25kW/m² heat flux would not be expected to fail within 15 minutes. By this time, ESD and blowdown will have been initiated, and the release rate will have been reduced such that the heat radiation at the adjacent equipment is below 25kW/m² and the likelihood of escalation reduced. Damage to cables etc. that will require a shutdown to replace may still occur.

Jet Fires

Representative hole sizes as shown in Table 1 – Modelling parameters of section 3.1 above, will be modelled and evaluated for jet fire and consequences. It is not considered practicable to specify separation distances based on jet fire sizes. This is due to the very low probability of a gas release leading to a jet fire that will impinge on neighbouring equipment (i.e. low risk) versus the very large separation distances which would be required if jet fires were the design case for separation (i.e. high cost).

In order to achieve a jet fire that can escalate to adjacent equipment, the following must occur:

- an ignition source
- stable ignition of the jet
- appropriate orientation of the release, with an unobstructed route

FRED models releases in unobstructed areas, under 'ideal' conditions, where a stable gas jet is formed. In reality high pressure releases, particularly from small hole sizes, will very often not be able to form a stable jet. Releases within modules are very likely to be obstructed, resulting in the gas jet losing momentum, giving a significantly smaller flame length than predicted. The results from FRED will therefore tend to over-state the hazard presented by jet fires.

The protection strategy against jet fires will therefore be in the prevention of releases through simple, robust plant design, and by prevention of ignition. Containment by isolation and blowdown will also play an important part, but may not be able to prevent escalation due to an impinging fire for all cases. Release orientation can also be controlled by location of fittings etc. such that releases will tend to be away from other equipment or adjacent inventories (i.e. identification of more likely release sources such as sight glasses, small bore line, and orientation of these such that any release has the minimal consequence).

Gas Dispersion

Representative hole sizes as shown in Table 1 – Modelling parameters of section 3.1 above, will be modelled and evaluated for gas dispersion and consequences. However, minimum separation distances between process equipment, control room, etc. will be specified based on the downwind LFL gas dispersion from a 25mm release, under 5D wind conditions as the design case.

3.2.3. Isolation & Blowdown

Isolation and blowdown are key controls in preventing escalation by controlling fire duration. Isolation valves will be located such that the inventory on the other side of the valve is not vulnerable to the fire - i.e. valves should be located at the edges of modules where the pipework on the other side of the valve will be less vulnerable to a fire from within the module.

The blowdown philosophy is to reduce pressure to 50% of design in 15 minutes or 7 bar whichever is lower. This is based on protection of a vessel from failure due to a fire at the vessel. This rate is not intended to prevent escalation to adjacent inventories.

3.2.4. Fire & Gas Detection

Where there are specific likely release sources, targeted fire and/or gas detection will be considered, and recommendations made where appropriate. Some types of equipment present both a higher release probability and a higher ignited release probability than other ones. There is therefore a greater potential benefit from installing fire & gas detection in these areas.

Fire and Gas Detection Philosophy usually specifies that gas detection be provided in order to warn personnel of accumulations of gas or large releases which will not rapidly disperse. Detection of small release in open modules will be difficult to achieve. This is due to the nature of gas releases on the site. They are generally of lighter than air gas, released at pressure and producing long, narrow gas plumes. This is accepted, as small releases are likely to disperse more rapidly and present a lower ignition and explosion hazard.

3.2.5. Passive Fire Protection

Applicability of PFP for jet fires and pool fires will be evaluated. Inventories will be checked as well for specific BLEVE scenarios and application of PFP as appropriate.

3.2.6. Active Fire Protection

Two types of response will be considered for the facilities under active fire protection, consistent with the Fire Protection philosophy of the project:

- Fixed fire-fighting and fire suppression systems (e.g. deluge, foam injection)
- Manual response by ER crews using mobile equipment.

The following assumptions will be made as to the effectiveness of active fire protection at mitigating the effects of different fire hazard scenarios.

Hazard	Fixed Systems	Manual Response
Gas Releases - unignited	General area water spray can be effective at dispersing gas clouds and preventing ignition. However, the nature of the releases at the manifolds means that large gas clouds are not expected and will not present a significant risk. This type of protection will therefore not be considered worthwhile.	No approach by personnel will be made to an unignited gas release.
Gas Release - ignited	Deluge is not effective against an impinging jet fire. Deluge is only effective against radiated heat (exposure protection). Given the low probability of achieving a sustained jet fire, and the fact that different inventory sections are separated and high-risk inventories have automatic detection, isolation and blowdown, deluge will not be considered worthwhile.	No attempt will be made to extinguish a gas fire. Water spray can be set up between equipment to prevent escalation due to radiated heat. This may not be possible or useful in many cases (due to direct impingement/direction of release).
Flashing Liquid Release - unignited	A flashing liquid release will be from pressure sources, to form a pool. The manifolds are located on swamp piles and therefore formation of liquid pools around the equipment is unlikely	No approach by personnel will be made to a flashing liquid release, due to the danger presented by the vapour.

Hazard	Fixed Systems	Manual Response
Flashing Liquid Release - ignited	A liquid release will be from pressure, and will burn at least partly as a spray fire. Extinguishment is therefore very difficult. Remotely operated monitors could be used to provide cooling of adjacent equipment - care would need to be taken to avoid spreading the pool.	For the project NAG inventories, the hazard presented by the gas inventory mean that it is not safe for ER personnel to approach the area.
Stable Liquid Release - unignited	Not applicable	Foam covering will help to prevent ignition.
Liquid Release - Ignited	Manifolds are on swamp piles, so it is unlikely that pools will form around the equipment. Therefore, deluge protection of vessels is not required/will not be cost-effective.	Foam can be used to extinguish a liquid fire. There will be limited cases in which this would reduce the consequences of a fire, however, as the nature of the facility (swamp piles) limits pool formation Manual firewater application can be used to reduce heat at adjacent equipment.

3.2.7. Bunding & Drainage

Drainage will be a key measure in controlling pool fire size, and preventing liquids gathering. Drainage should be specified to prevent liquids gathering under vulnerable equipment, and to prevent liquids travelling from one area/train to another.

The manifolds are on swamp piles, so running pool will be on the water surface and away from the equipment on the platform. In order to limit environmental impact, any areas where leaks are likely should be provided with drip pans and channelled to the open drain system

3.2.8. Explosion Assessment

This will be done using the Congestion Assessment Module (CAM) in FRED, to model explosion overpressures based on the level of congestion within the project facilities

3.3. Risk Analysis & Acceptance Criteria

Risk analysis and acceptance criteria will be carried out in line with Shell ALARP Guide